Appendix

PROJECT 1 SUPPORTING TABLES AND EXTENDED FIGURES

This appendix contains detailed diagnostics and overflow visual content not included in the main 12-page `Report.pdf`.

A. EXTENDED AUDIT TABLES

A1. Duplicate and Key-Integrity Audit

Table	Duplicate rows	Row count
train	0	307,511
test	0	48,744
app	0	356,255
bureau	0	1,716,428
bureau_agg_all	0	305,811
prev_agg	0	338,857
pos_agg	0	337,252
cc_agg	0	103,558
ins_agg	0	339,587

A2. Top Missing Columns (Train-Join Context)

Column (sample)	Missing rate
PREV_APP_RATE_INTEREST_PRIMARY_MEAN	0.985012
PREV_APP_RATE_INTEREST_PRIMARY_MAX	0.985012
PREV_APP_RATE_INTEREST_PRIMARY_MIN	0.985012
PREV_APP_RATE_INTEREST_PRIVILEGED_MEAN	0.985012
PREV_APP_RATE_INTEREST_PRIVILEGED_MAX	0.985012
PREV_APP_RATE_INTEREST_PRIVILEGED_MIN	0.985012
CREDIT_CARD_AMT_PAYMENT_CURRENT_MEAN	0.801438
CREDIT_CARD_AMT_PAYMENT_CURRENT_MAX	0.801438
CREDIT_CARD_AMT_PAYMENT_CURRENT_MIN	0.801438
CREDIT_CARD_AMT_DRAWINGS_POS_CURRENT_MEAN	0.801178

A3. Missing-Rate Bucket Summary (Train Fold)

Missing-rate bucket	Feature count	Avg missing rate
0-5%	63	0.000325
5-20%	590	0.073462
20-60%	42	0.469369
60-100%	196	0.725647

A4. Imputation Strategy Scoreboard (Validation ROC-AUC)

Rank	Imputation type	Val AUC
1	OCCUPATION_TYPE + Median	0.701886
2	CODE_GENDER + Median	0.700218
3	NAME_INCOME_TYPE + Median	0.699069
4	OCCUPATION_TYPE + Mean	0.697287
5	Global_Median_Baseline (No Grouping)	0.696289
6	CODE_GENDER + Mean	0.696242
7	NAME_EDUCATION_TYPE + Median	0.694470
8	NAME_INCOME_TYPE + Mean	0.694338
9	NAME_EDUCATION_TYPE + Mean	0.692746

A5. Winsorization Grid Search (Validation ROC-AUC)

lower_q	upper_q	val_auc	train_cell_change_rate
0.0500	0.9500	0.708085	0.080050
0.0000	0.9500	0.706108	0.042151
0.0050	0.9800	0.705814	0.020611
0.0200	0.9800	0.705748	0.031799
0.0100	0.9900	0.705214	0.015988
0.0010	0.9900	0.704914	0.009253
0.0000	0.9800	0.704590	0.016865
0.0000	0.9900	0.704141	0.008597
0.0050	0.9950	0.703755	0.007976
0.0025	0.9975	0.703517	0.003760
0.0000	0.9950	0.702759	0.004231
0.0010	0.9990	0.701909	0.001481
0.0000	0.9990	0.701148	0.000826
0.0100	1.0000	0.693749	0.007391
0.0500	1.0000	0.692577	0.037898

A6. Drift Reports Used for Feature Filtering

Numeric drift (top sample)

Feature	PSI	KS stat	Train mean	Test mean
BUREAU_BUREAU_BALANCE_ROW_COUNT_MEAN	0.313084	0.245179	27.477237	36.304418
BUREAU_BUREAU_BALANCE_MONTHS_BALANCE_MEAN_MEAN	0.171562	0.150472	-20.984805	-17.683255
BUREAU_DAYS_CREDIT_MIN	0.029766	0.045133	-1762.374882	-1776.740217

Categorical drift (top sample)

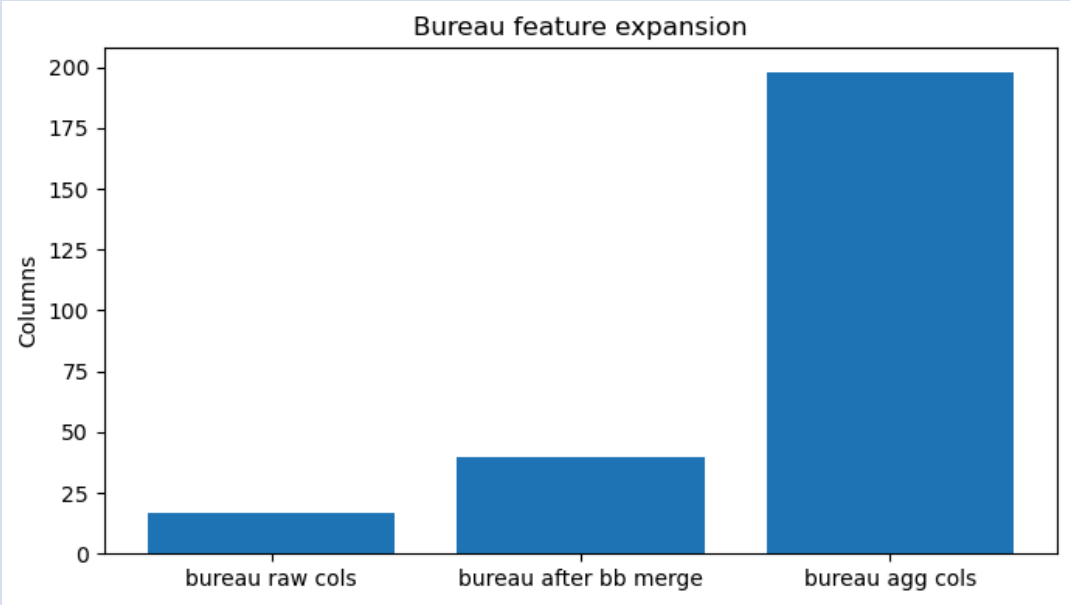
Feature	L1 share shift	Max category shift
NAME_CONTRACT_TYPE	0.172413	0.086207
WEEKDAY_APPR_PROCESS_START	0.090183	0.024764
NAME_FAMILY_STATUS	0.047033	0.023516

A7. Feature-Drop Summary by Stage

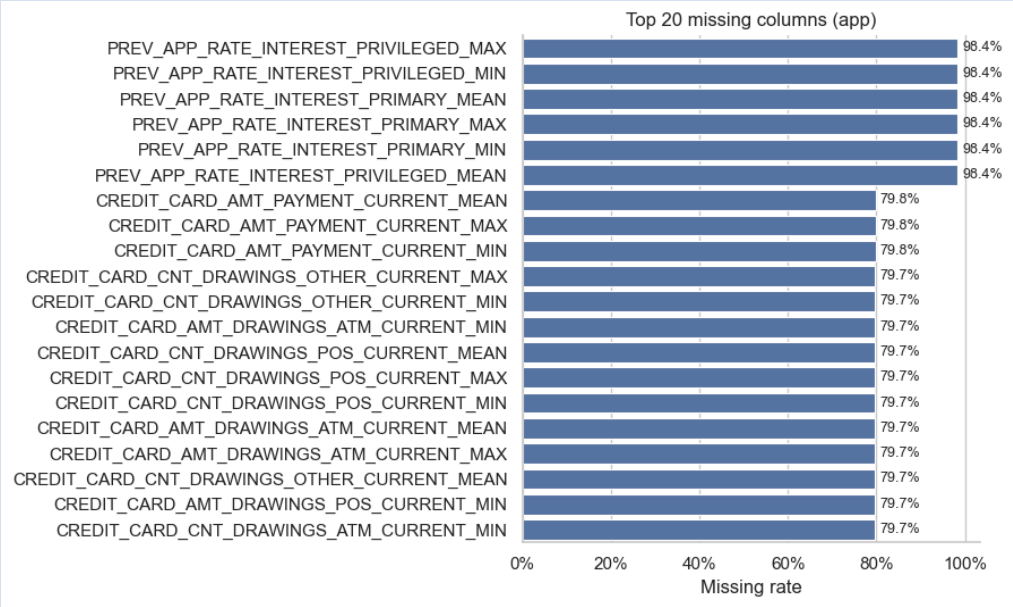
Stage	Features removed	Remaining
High missing removal	196	695
Data drift removal	3	694
Low-information removal	259	438
High-correlation removal	42	396
Total removed	**500 (stage sum, includes intermediate overlaps and additions)**	**396 final**

Note: Main report tracks net reduction from pre-pruning to final matrix as **495** columns.

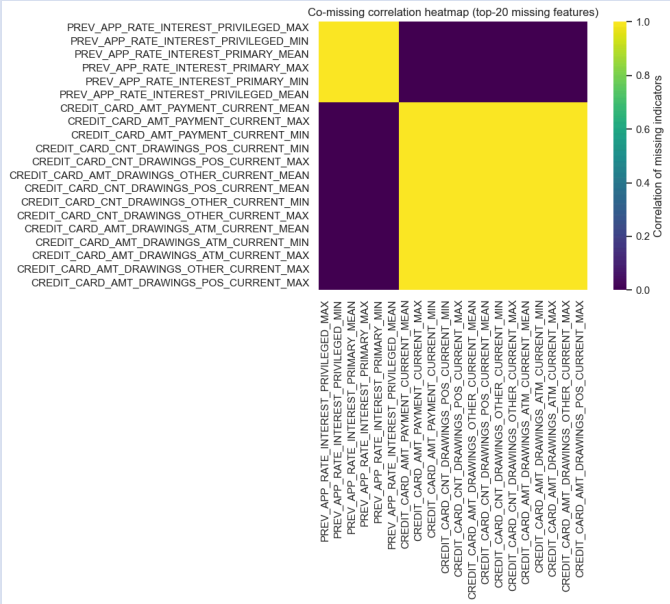
B. EXTENDED FIGURE GALLERY



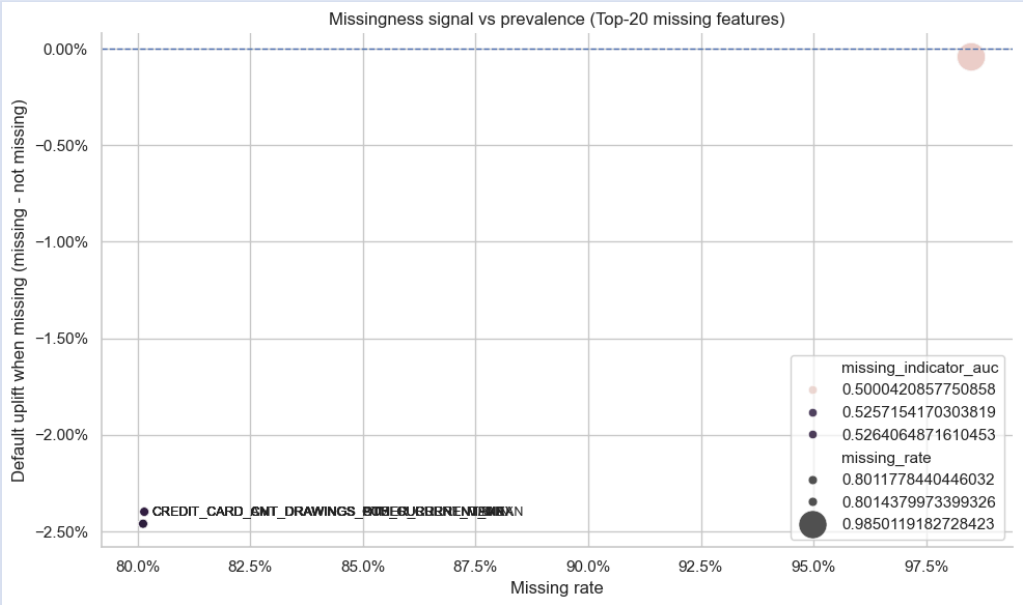
Appendix Figure A1: Bureau table feature expansion after bureau_balance enrichment and aggregation.



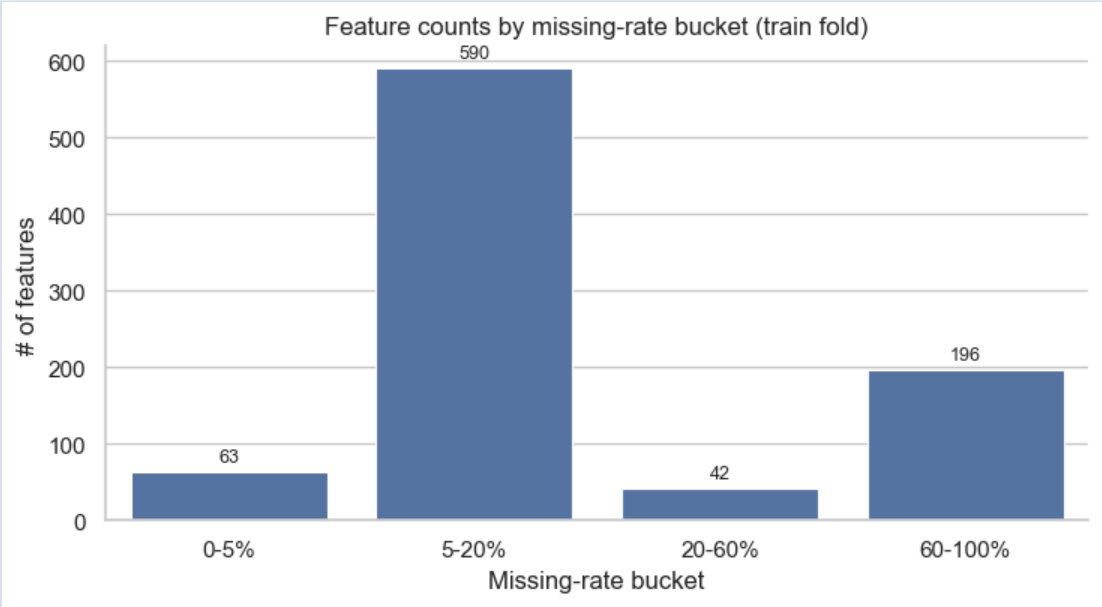
Appendix Figure A2: Top 20 columns by missing rate in application-level joined data.



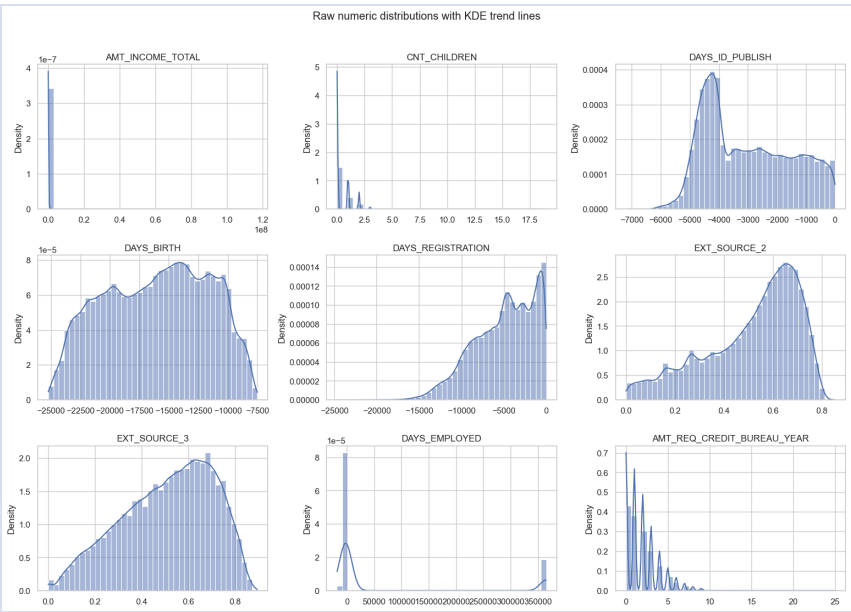
Appendix Figure A3: Correlation structure among missingness indicators for top-missing features.



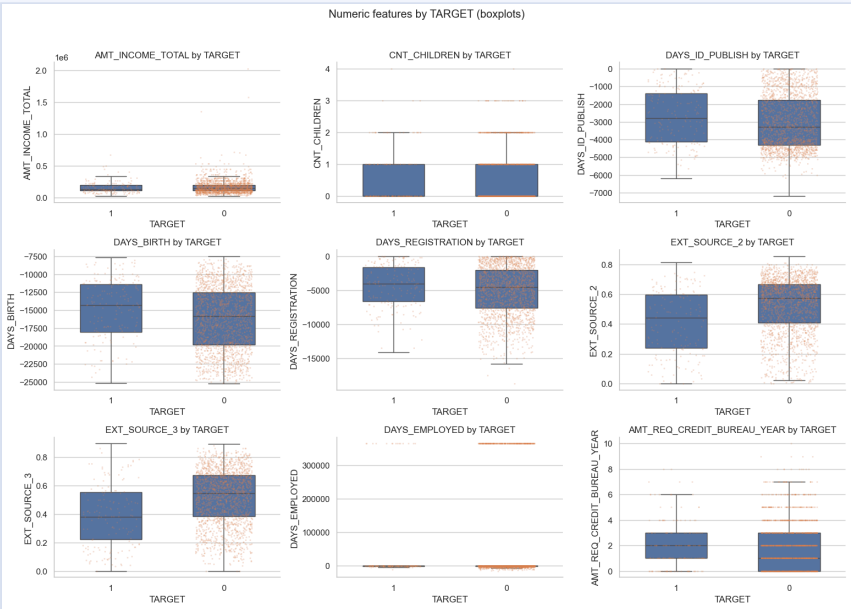
Appendix Figure A4: Missingness prevalence vs default uplift diagnostic plot.



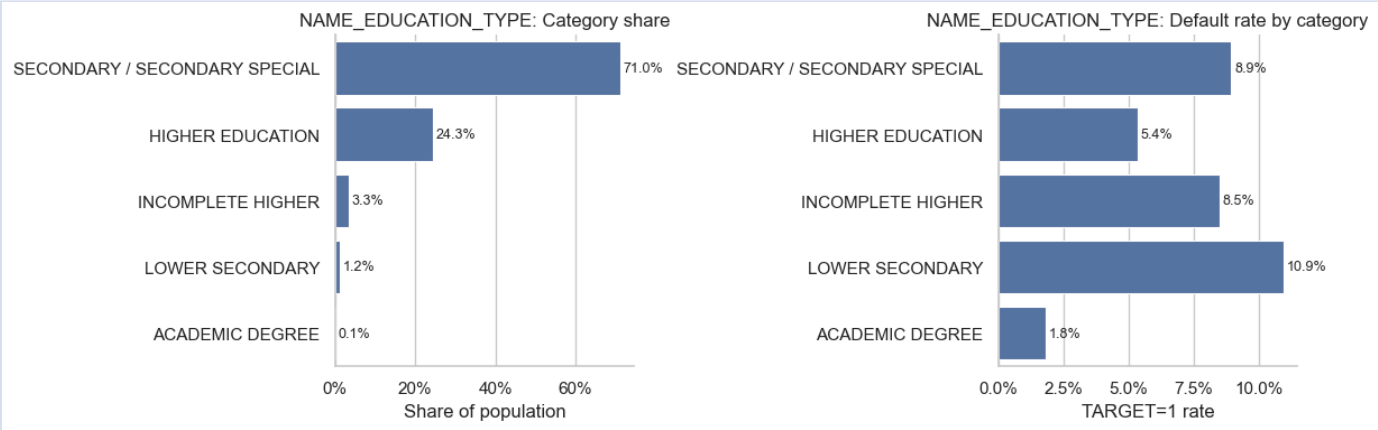
Appendix Figure A5: Count of features by missing-rate bucket on training fold.



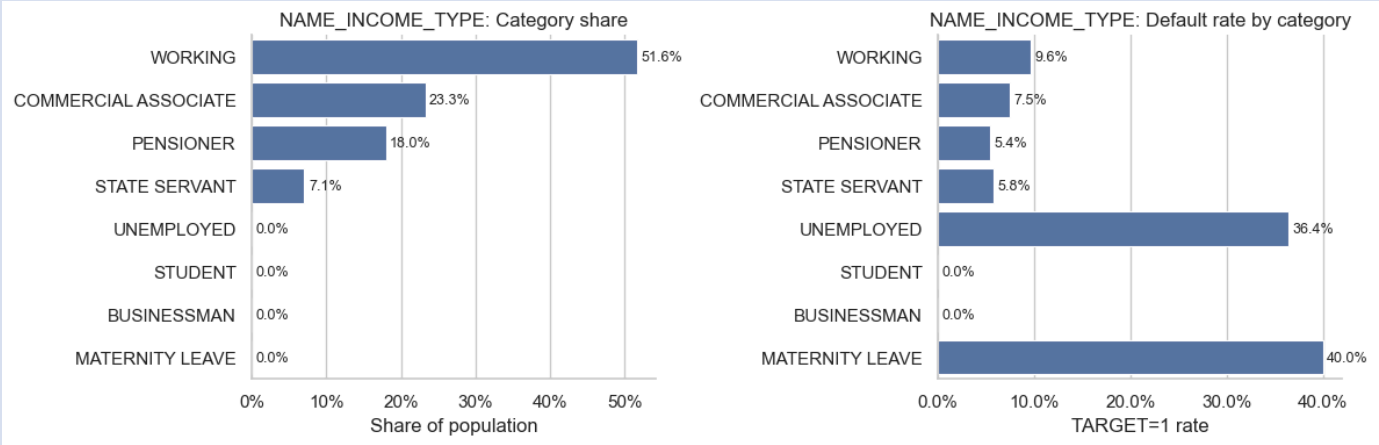
Appendix Figure A6: Raw numeric distributions before deep preprocessing.



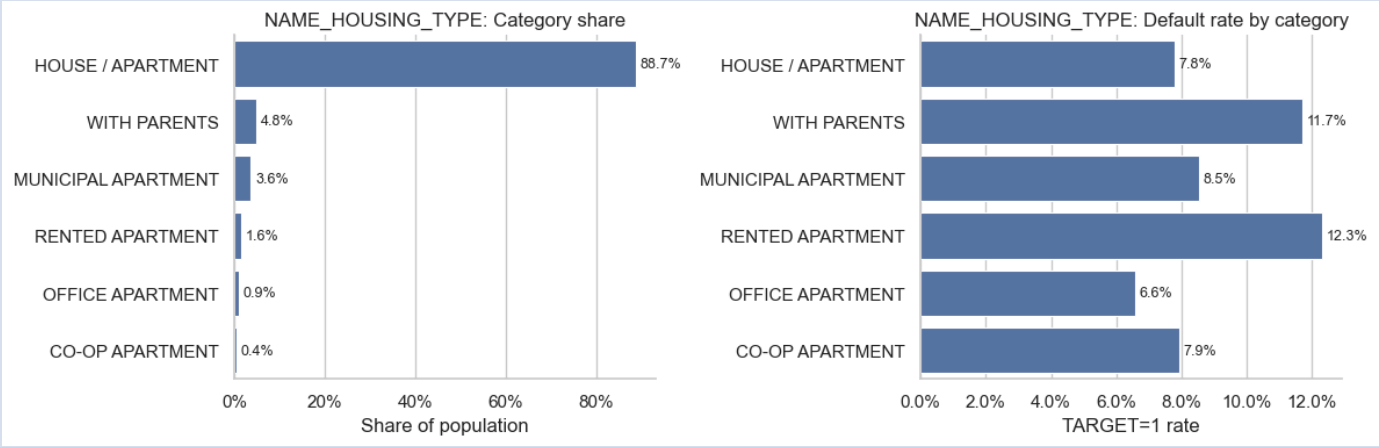
Appendix Figure A7: Numeric feature distributions by class label.



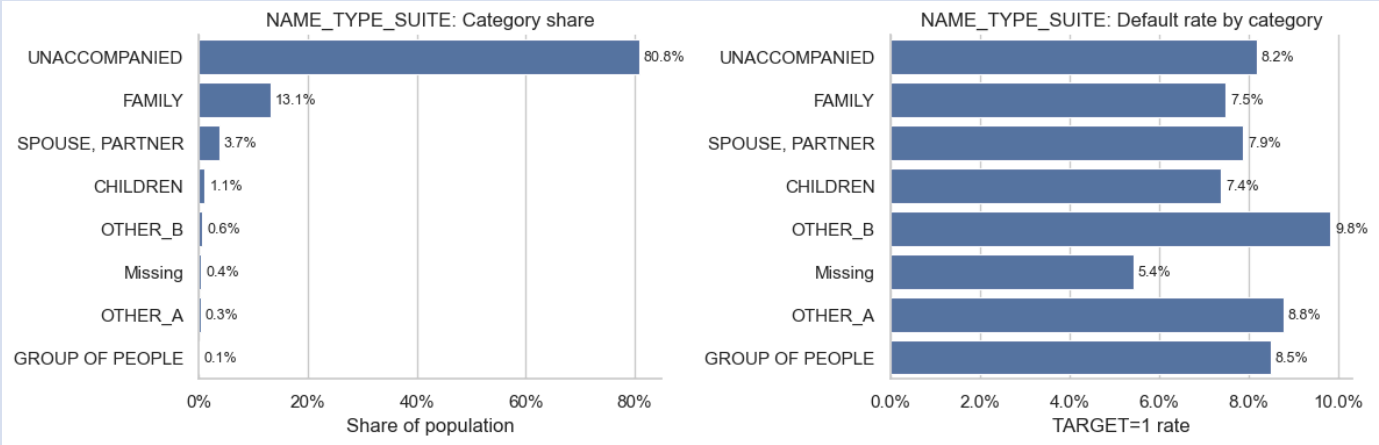
Appendix Figure A8: Default-rate profile by education type.



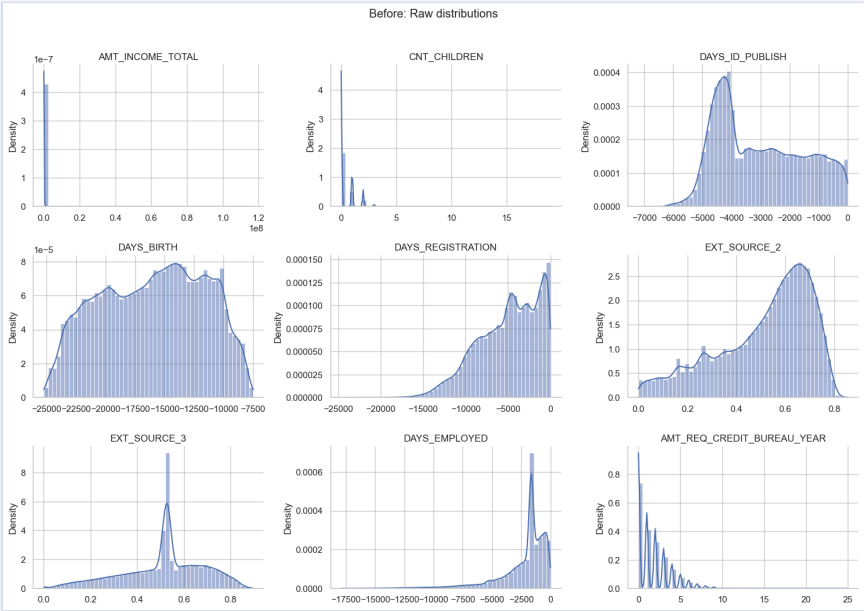
Appendix Figure A9: Default-rate profile by income type.



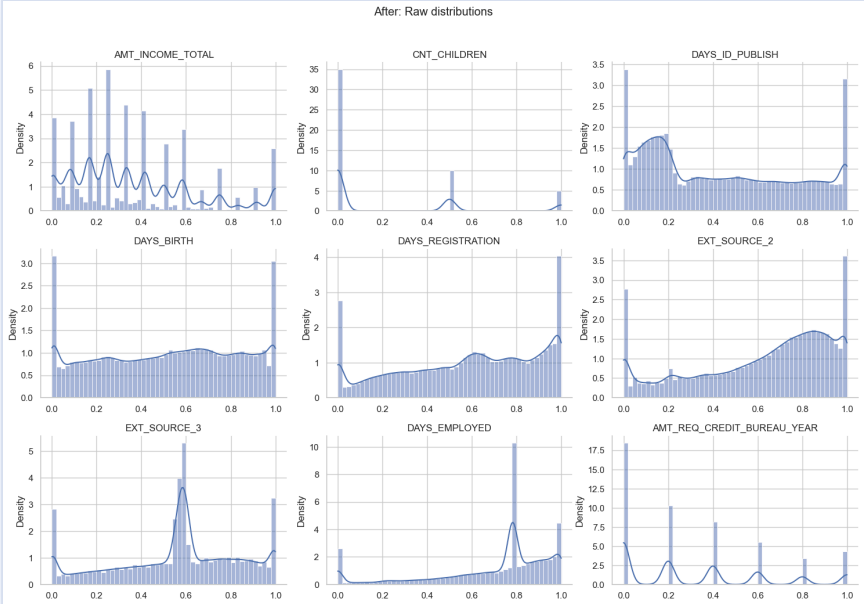
Appendix Figure A10: Default-rate profile by housing type.



Appendix Figure A11: Default-rate profile by suite type.



Appendix Figure A12: Representative distribution panel before late-stage processing.



Appendix Figure A13: Representative distribution panel after preprocessing and resampling.

For full run details, refer to `Project_1.ipynb` and CSV logs under `log/`.